TECH GEEKS

ROUND 1: IDEATION

Present your idea, problem statement, proposed solution, and project approach through a PPT

ROUND 2: IMPLEMENTATION

Build your solution, implement key features, and demonstrate a working prototype

LinkPulse: URL Shortener & Analytics Platform

An Integrated Link Management, Redirection, and Analytics Platform
This hackathon project is a URL shortening platform called “LinkPulse”, designed to handle:

- Short link creation with auto-generated or custom aliases, expiry rules, and password protection
- Fast, reliable redirection with rules for link status, expiry, click limits, and device targeting
- Click tracking that captures time, referrer, device, browser, and location for every visit
- Link organization through workspaces, folders, tags, and campaigns
- QR code generation, bulk link import, and link-in-bio style landing pages
- A live Analytics Dashboard that combines link, click, campaign, and abuse-monitoring data

Many basic URL shorteners only map a short code to a long URL. Real marketing and operations teams need links, analytics, and governance to work together. A single link may belong to a campaign, expire next month, redirect mobile users to a different destination, and need to be disabled quickly if it turns out to be malicious. Click data must be recorded without slowing down redirects, and the dashboard must turn that raw data into insights people can act on.



Project Overview

The goal is to build a URL shortener that goes beyond “paste a URL, get a short URL”. The Link record is the central hub. Campaigns provide organization, Redirect Rules control routing, Click Events capture every visit, and the Dashboard turns that data into reports.
Teams are free to use any language, framework, or database. The focus is on business logic, data relationships, redirect flow, and analytics accuracy, not on any specific platform.

Main Goal

Build an integrated platform that manages the full link lifecycle: creation, redirection, tracking, analytics, and expiry or takedown.

Key Outcomes

- **Unified Link Flow:** Central link records with navigation to Campaigns, Clicks, and QR Codes.
- **Reliable Redirection:** Unique short codes, redirecting only when the link is active, unexpired, and within its click limit.
- **Basic Rules:** Support expiry dates, click caps, and password protection.
- **Link Creation Workflow:** A two-step flow: enter the destination and alias, then set options and review validation warnings.
- **Reporting:** A dashboard aggregating click and link data by Period and Campaign.

Goals & Scope

- Every redirect, rule, and report is driven by real data, not hardcoded values
- Redirects stay fast while clicks are still recorded reliably
- Problems such as duplicate aliases and invalid URLs are surfaced before publishing
- Expired or disabled links remain traceable in reports



User Roles

Visitor (Anonymous)

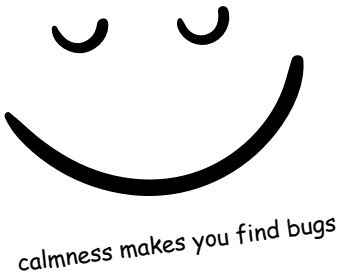
- Open short links and enter passwords
- No management access

User

- Manage own links, campaigns, tags, and QR codes
- View analytics for own links only

Admin

- Full access to all links, users, and settings
- Disable abusive links and manage the domain blocklist
- View platform-wide analytics



Features

Management Backend

- **Link Management:** Create, edit, organize, and manage short links with custom aliases, tags, campaigns, and expiry dates.
- **Redirect Rules:** Set expiry dates, click limits, and passwords to control who can access a link.
- **Campaigns & Tags:** Group links by campaign or tags and view total links, clicks, and top-performing links.
- **Click Tracking:** Track when and where people click, their devices and browsers, and unique visitors while filtering bots.
- **Safety Checks:** Validate URLs, block unsafe websites, and let admins disable suspicious links.

Platform Frontend

- **Navigation & Link Details:** Manage links, campaigns, QR codes, analytics, and settings in one place.
- **Link Creation:** Create short links, configure rules, validate URLs, and generate QR codes.
- **Bulk Import:** Upload CSV files, identify errors and duplicates, and import valid links.
- **Redirect Engine:** Check link status, expiry, click limits, and passwords before redirecting.
- **QR Codes:** Generate downloadable QR codes and track scans.
- **Link Analytics:** Track clicks, unique visitors, traffic sources, devices, and browsers.
- **Analytics Dashboard:** View key metrics, campaign performance, top links, and link alerts.

Complete Flow

- Users create campaigns and tags
- Users create links via the wizard with validation
- Links are shared as short URLs or QR codes
- Visitors open a link; the engine checks status, expiry, limits, and password
- Each visit is logged, with bots filtered out
- Users review link and campaign analytics
- Admins disable abusive links, stopping redirects immediately
- Expired or disabled links keep their analytics

Deliverables

- **Functional platform:** Working shortener with sample links, campaigns, and clicks
- **Live demo:** Five-minute walkthrough of two scenarios, e.g., link creation to analytics, and bulk import to link disable
- **Future roadmap:** Short list of next steps (e.g., custom domains, device targeting, public API)

Evaluation Focus

- **Unified Workflow:** Links, campaigns, rules, clicks, and analytics working as one flow.
- Business Logic: Code generation, rule order, click limits, bot filtering.
- **Architecture:** Role-based access, async processing, clear data relationships.
- **Versatility:** Any stack, with priority on solid data modeling and accurate behavior.

We do not expect teams to implement every feature listed above. At a minimum, we want a functional platform with a hosted link: users should be able to create a short link, be redirected through it, and see basic click tracking. Prioritize a working, deployed core over a long list of half-built features.



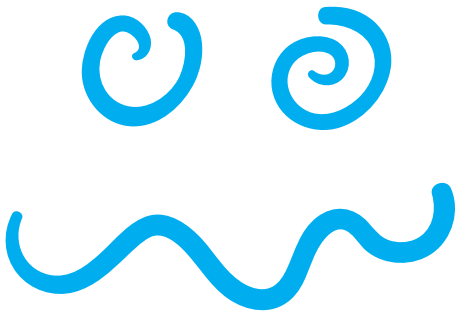
APART FROM PROBLEM STATEMENT

Why This Problem Is Important?

- Combines creation, redirection, tracking, and safety in one flow
- Tests real logic: unique codes, rule order, click limits, bot filtering
- Encourages role-based access, caching, and async click logging

Technical Guidelines

- Any backend, frontend, and database
- Build code generation, alias validation, and rule evaluation in application logic
- Rules must drive actual redirect behavior
- Show alias and URL issues before publishing
- Dashboard must use live data
- Include QR generation and CSV import
- Apply basic rate limiting on link creation



We're excited to see your submissions!